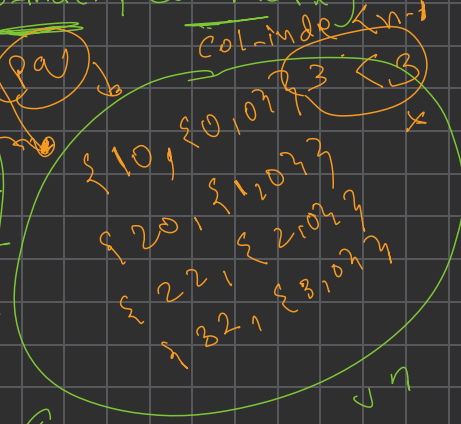




Pair $\rightarrow$ (value, row-index, col-index)

$10 \rightarrow 13$
 $15 \rightarrow 13$
 $20 \rightarrow 13$
 $25 \rightarrow 13$
 $30 \rightarrow 13$

	0	1	2	3
0	10	15	20	25
1	20	25	30	35
2	22	30	37	48
3	32	33	39	50



$(20) \rightarrow (20, (1, 0))$
 Min
 $22, 32$
 30

$O(n)$

$K \times \log n$

$n + K \log n$

$pair < int, pair < int, int > >$
 $\Downarrow$ $\Downarrow$ $\Downarrow$
value row-index col-index

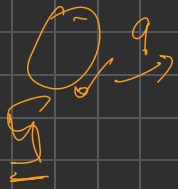
10 ✓	20 ✓	30	40
15 ✓	25 ✓	35	45 ✓
27 ✓	29 ✓	37 ✓	48 ✓
32 ✓	33	39	50

7th smallest

$O(K)$

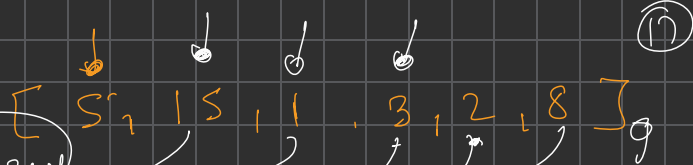
$(n^2 - K) + \log K$
 $n^2 \log K$

10 20 30 32
 15 25 29



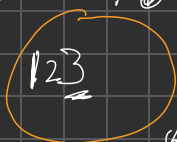
Left side > Right side
 Max heap Min heap

$O(n^2)$

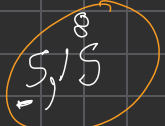


median

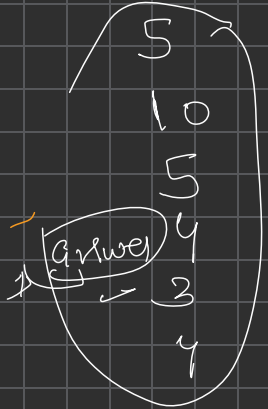
problem



Max Left



Min heap R.H



$$\frac{18 + 19}{2} \rightarrow 18.5$$

12

